

$$\begin{array}{ccc}
\iota A \otimes_C \iota B & \xrightarrow{\iota \operatorname{coev}_A^B \otimes_C \iota B} & \iota [B, A \otimes_D B] \otimes_C \iota B \\
\downarrow \operatorname{coev}_{\iota A}^{\iota B} \otimes_C \iota B & & \downarrow \psi_{B, A \otimes_D B} \otimes_C \iota B \\
[\iota B, \iota A \otimes_C \iota B] \otimes_C \iota B & \xrightarrow{[\iota B, \eta_{\iota A \otimes_C \iota B}] \otimes_C \iota B} [\iota B, \iota F(\iota A \otimes_C \iota B)] \otimes_C \iota B & \xlongequal{\quad} [\iota B, \iota (A \otimes_D B)] \otimes_C \iota B \\
\downarrow \operatorname{ev}_{\iota A \otimes_C \iota B}^{\iota B} & \downarrow \operatorname{ev}_{\iota F(\iota A \otimes_C \iota B)}^{\iota B} & \downarrow \operatorname{ev}_{\iota (A \otimes_D B)}^{\iota B} \\
\iota A \otimes_C \iota B & \xrightarrow{\eta_{\iota A \otimes_C \iota B}} \iota F(\iota A \otimes_C \iota B) & \xlongequal{\quad} \iota (A \otimes_D B)
\end{array}$$